Endocrine Cholestasis: Septo-Optic Dysplasia, Hypopituitarism, Hypothyroidism and Cortisol Deficiency in the Jaundiced Infant.

Picture two infants referred to the liver clinic in the same week. The first is a three-week-old boy whose direct bilirubin is five milligrams per deciliter and whose gamma-glutamyl transferase, the liver enzyme usually shortened to GGT, is sixty units per liter. His newborn screen was normal. On the second day of life he had a low blood sugar that the nursery called transitional and forgot. When you examine him carefully, his penis is small and his eyes wander without fixing on your face. The second is a seven-week-old girl sent in for a biliary atresia work-up because her stools look pale. She has had several low sugars, her GGT is normal, and the referring team has already sent a genetic panel for progressive familial intrahepatic cholestasis. Both infants have a hormone problem, not a primary liver disease, and in both the hormone problem is the thing that could kill them.

The clinical question of this chapter is when the liver is the first organ to signal an endocrine disorder, and how to find that disorder quickly. The main idea is this. Endocrine cholestasis is rare in the liver clinic, about one infant in fifty, but it is common among infants born with a failing pituitary gland, and it is driven above all by a lack of cortisol. Cortisol keeps the bile acid pumps of the liver cell switched on. When cortisol is missing, bile flow falls, conjugated bilirubin rises, and the GGT may be normal or raised, so the picture can mimic either a genetic transporter disease or biliary atresia. The cholestasis goes away with hormone replacement over weeks to months. The real danger is not the liver at all. It is adrenal crisis, which can cause sudden death, and which can be provoked by giving thyroid hormone before cortisol.

We will move through the topic in order. First, the mechanism by which hormones control bile flow. Second, how often an endocrine cause explains infant cholestasis. Third, what the affected infant looks like, including the trap of the GGT. Fourth, the evidence that cortisol is the main driver, and the bedside clues. Fifth, how to test, including the blind spot of the newborn screen. Sixth, treatment and outcome, and where primary hypothyroidism fits. We then compare the guidelines, list the traps, and finish with what to do differently in clinic.

A note on sources. The central hepatology guideline is the 2017 guideline on the evaluation of cholestatic jaundice in infants, written by Fawaz and colleagues for the North American Society for Pediatric Gastroenterology, Hepatology and Nutrition, known as NASPGHAN, and the European Society for Paediatric Gastroenterology, Hepatology and Nutrition, known as ESPGHAN. For frequency we use the 2015 systematic review by Gottesman and colleagues in BMC Pediatrics, which pooled 1,692 infants with conjugated hyperbilirubinemia. For the clinical picture we use three case series: Braslavsky and colleagues from Buenos Aires in

2011, Grammatikopoulos and colleagues from King's College Hospital in London in 2018, and Karnsakul and colleagues from Thailand in 2007, together with a French cohort published by Mauvais and colleagues in 2016. For thyroid screening we use the Dutch data of van Tijn and colleagues from 2005 and Lauffer and colleagues from 2021, and for treatment we use the 2021 congenital hypothyroidism consensus guideline of the European Society for Paediatric Endocrinology, led by van Trotsenburg.

Part one. How hormones keep the bile pumps running.

To understand why a missing hormone makes an infant yellow, start with how bile is made. Bile acids are taken up from the portal blood into the liver cell, the hepatocyte, by uptake transporters on the side of the cell that faces the blood. The most important is the sodium taurocholate cotransporting polypeptide, called NTCP. Bile acids then leave the cell into the tiny bile channel between neighboring cells, the canaliculus, through the bile salt export pump, called BSEP. Phospholipid is flipped into bile by a second pump, called MDR3, and conjugated bilirubin is exported by a third, called MRP2. The movement of bile acids into the canaliculus pulls water after them, and that is bile flow. A second part of bile flow does not depend on bile acids at all and depends instead on sodium and other ions, driven by the sodium potassium pump in the cell membrane.

Cortisol acts through the glucocorticoid receptor, a protein inside the hepatocyte that, once bound to cortisol, moves to the nucleus and switches genes on. The transporter genes are among its targets. In cultured rat liver cells, the synthetic steroid dexamethasone raised the messenger RNA for the bile salt export pump about tenfold, according to work by Warskulat and colleagues published in 1999. In mice, Rose and colleagues showed in 2011 that knocking down the glucocorticoid receptor only in the liver reduced the uptake transporter NTCP and impaired bile acid uptake. The human evidence comes from the King's College series: in infants with congenital hypopituitarism who had liver biopsies, staining for the bile salt export pump and for MDR3 along the canaliculus was reduced, and MRP2 was reduced in two. In other words, an infant without cortisol develops an acquired, reversible version of the transporter failure that is permanent in the genetic cholestasis syndromes.

Thyroid hormone works on the other part of bile flow. In a classic experiment published by Layden and Boyer in the Journal of Clinical Investigation in 1976, rats made hypothyroid lost about half of the activity of the sodium potassium pump on the canalicular membrane, and their bile acid independent bile flow fell. When thyroxine was given, flow recovered within two days. Growth hormone has a supporting role. Its signaling protein in the liver, called STAT5, physically interacts with the glucocorticoid receptor, and mice lacking both develop fatty liver and liver cancer. But the evidence that growth hormone deficiency on its own causes neonatal cholestasis is indirect, and the clinical data, as we will see, point to cortisol as the

dominant driver.

This mechanism explains the most confusing feature of endocrine cholestasis, the GGT. GGT sits on the canalicular membrane and is released into the blood when bile acids in the canaliculus act on that membrane. When the bile salt export pump is switched down, fewer bile acids reach the canaliculus, so GGT may stay normal, exactly as in the genetic disorders of the pump. When inflammation and giant cell change dominate the biopsy, the GGT rises instead. Both patterns occur in the same disease, sometimes in the same child at different times.

One common teaching should be dropped. It is often said that hypothyroidism causes jaundice by lowering the bilirubin conjugating enzyme, uridine diphosphate glucuronosyltransferase 1A1. In hypothyroid rats, Visser and colleagues found in 1996 that the activity of this enzyme actually rose. The prolonged unconjugated jaundice of hypothyroid infants is better explained by slower gut transit, which returns more bilirubin to the blood through the enterohepatic circulation, and by reduced uptake and flow.

Part two. How often is it endocrine?

In the systematic review by Gottesman and colleagues, the combined category of hypopituitarism and hypothyroidism accounted for thirty-three of 1,692 cholestatic infants, or just under two percent. For comparison, idiopathic neonatal hepatitis accounted for twenty-six percent, biliary atresia for just under twenty-six percent, infection for eleven and a half percent, parenteral nutrition for about six percent, metabolic disease for about four and a half percent, alpha one antitrypsin deficiency for about four percent, and progressive familial intrahepatic cholestasis and Alagille syndrome for about one percent each. The review did not count adrenal causes separately. In a Turkish series of 131 cholestatic infants published in 2021, endocrine disease explained five point three percent: five infants with congenital hypothyroidism, one with hypopituitarism and one with adrenal insufficiency.

Now look from the other side. Among infants born with a failing pituitary gland, cholestasis is common. In the French cohort of Mauvais and colleagues, five of sixteen infants with pituitary stalk interruption syndrome diagnosed before their first birthday were cholestatic. A 2025 review that pooled the published stalk interruption series found cholestasis in between six and forty-two percent. So both statements are true at once: the liver clinic sees an endocrine cause in about one infant in fifty, while the endocrine clinic sees cholestasis in up to four in ten infants with neonatal-onset hypopituitarism. Low prevalence in the liver clinic is not a reason to skip a cheap blood test, because the cause is fully treatable and the untreated adrenal axis can kill.

Part three. What the affected infant looks like.

The Buenos Aires series by Braslavsky and colleagues described sixteen cholestatic infants: eleven with multiple pituitary hormone deficiency and five with isolated central hypocortisolism, meaning cortisol deficiency caused by a lack of the pituitary hormone corticotropin, also known as ACTH. Cholestasis began at a median of eighteen days of age, with a range from two days to four months. The median direct bilirubin was four point eight five milligrams per deciliter, with a range from just over one to more than twenty. The GGT was raised in nine of sixteen, and the aminotransferases were raised in ten of sixteen, at two to ten times the upper limit of normal. Fourteen of the sixteen had spontaneous hypoglycemia, making a low blood sugar the most consistent clue in the whole series. And the referral to endocrinology took a median of thirty-two days. That month is the window in which an adrenal crisis can happen.

After replacement, cholestasis remitted at a median of sixty-five days, ranging from about three weeks to about five and a half months, and it remitted faster in the infants with isolated cortisol deficiency than in those with multiple deficiencies. The enzymes normalized at a median of ninety days. At a median follow-up of almost four years, no child had permanent liver dysfunction.

The King's College series by Grammatikopoulos and colleagues covered twenty-six years and included twenty-one infants with congenital hypopituitarism and eighteen with septo-optic dysplasia, the developmental disorder that combines underdevelopment of the optic nerves, pituitary hormone deficiency and midline brain defects such as an absent septum pellucidum. These infants reached the liver unit at a median of eight weeks. The median total bilirubin was 116 micromoles per liter, about seven milligrams per deciliter. The median GGT was ninety-five units per liter, with a very wide range from twenty-five to more than seven hundred, and six of the ten infants who had a biopsy had a normal GGT. The thyroid tests are instructive: the median free thyroxine was low, at nine point six picomoles per liter, but the median thyroid stimulating hormone, or TSH, was five point nine five milliunits per liter. In other words, this was central hypothyroidism, in which the pituitary cannot raise TSH, yet the TSH was not low. It sat in the normal or mildly raised range. The biopsies showed cholestasis within the lobule, non-specific hepatitis, giant cell transformation of hepatocytes and, in some, fibrosis. Staining for the pumps was reduced, as described earlier, and none of the six children who were sequenced carried a variant in the genes for the bile salt export pump or for familial intrahepatic cholestasis type one. Liver tests normalized at a median of six months, with a range of three to nine months, and the prognosis was good.

This brings us to the GGT trap. Across series, roughly half of cholestatic hypopituitary infants have a normal GGT: forty-four percent in Buenos Aires, sixty percent of the biopsied infants in London, and between forty-three and fifty-seven percent in the pooled stalk interruption series. In the Thai series the GGT was raised in every case, with a mean of 411 units per liter. So

GGT cannot exclude an endocrine cause in either direction. In the low GGT infant, the differential diagnosis includes progressive familial intrahepatic cholestasis, bile acid synthesis defects, the arthrogryposis, renal dysfunction and cholestasis syndrome, and endocrine cholestasis, and free thyroxine, TSH and cortisol should be measured before the genetic results return. In the high GGT infant, endocrine cholestasis can sit in the biliary atresia range, so a pale stool still mandates the atresia pathway on schedule. One recent case report even showed the GGT rising from twenty to 131 units per liter five days after hydrocortisone was started, presumably because the pumps came back on and bile acids again reached the canalicular membrane.

Part four. Cortisol is the driver, and the bedside clues.

Several lines of clinical evidence point to cortisol rather than thyroid hormone or growth hormone as the main cause. In the French cohort, the median plasma cortisol in the cholestatic infants was twelve point four nanograms per milliliter, against seventy-nine point four in the infants without cholestasis, a difference that was statistically significant. The cholestatic infants also more often had eye malformations, five of five against four of eleven, and more often had a low Apgar score. Isolated cortisol deficiency, without any thyroid or growth hormone problem, is enough to cause cholestasis: a Saudi report described four such infants who cleared their jaundice in about three months on hydrocortisone alone. And in Buenos Aires, as we saw, isolated hypocortisolism remitted faster than multiple deficiency.

The bedside clues follow from the anatomy. Think of the sugar, the genitals and the eyes. Cortisol and growth hormone both defend the blood glucose in the newborn, so hypoglycemia is the commonest sign: it was present in fourteen of sixteen infants in Buenos Aires and in eight of eight in Thailand. It is often labeled transitional on day two and forgotten by week four, so read the glucose chart. The genitals reflect the gonadotropins. A micropenis, defined at term as a stretched penile length under two and a half centimeters, and undescended testes point to central hypogonadism, and in the first months of life, during the hormonal surge called mini-puberty, a low luteinizing hormone with a low testosterone confirms it. The eyes reflect septo-optic dysplasia: roving nystagmus, failure to fix and follow, and optic nerve hypoplasia on fundoscopy. Look also for other midline defects, such as cleft palate or a single central incisor. A low Apgar score, breech birth, a sepsis-like collapse and low sodium complete the picture. Girls lack the genital clue, so in them the sugar and the eyes carry the diagnosis.

How common is septo-optic dysplasia? Review articles quote about one in ten thousand births. The European congenital anomaly registry, EUROCAT, counted a lower figure, between one point nine and two and a half per hundred thousand births. Either way, it is uncommon, which is exactly why a cholestatic infant with hypoglycemia should make you think of it.

Part five. Testing, and the newborn screen blind spot.

Many clinicians assume that a normal newborn screen excludes hypothyroidism. It excludes primary hypothyroidism, in which the thyroid gland fails and TSH rises. It does not reliably exclude central hypothyroidism, in which the pituitary cannot make TSH, so TSH stays normal or only mildly raised while thyroxine is low. Most American states and the United Kingdom screen primarily on TSH. The Netherlands screens with a combination of thyroxine, thyroxine binding globulin and TSH, and that program shows what TSH screening misses. In the analysis by van Tijn and colleagues, central congenital hypothyroidism was found in about one in twenty thousand infants, later estimated at about one in sixteen thousand four hundred. In that series, seventy-eight percent had multiple pituitary hormone deficiencies, and mortality reached as high as fourteen percent when detection was late. Lauffer and colleagues later estimated the true incidence at about one in thirteen thousand once missed cases were added, and found that sixty-one percent of screen-detected central hypothyroidism was part of multiple pituitary deficiency. Even the Dutch program screens preterm infants on TSH alone.

The 2017 NASPGHAN and ESPGHAN guideline asks the clinician, in the first tier of tests, to verify the newborn screen results for treatable disorders such as hypothyroidism and galactosemia, and to check TSH and thyroxine if the screen results are not readily available. It places TSH, thyroxine and cortisol in the second tier. For the hepatologist, the practical rule is stronger: measure free thyroxine and TSH together in every cholestatic infant, and read free thyroxine first. A low free thyroxine with a high TSH is primary hypothyroidism. A low free thyroxine with a low, normal or mildly raised TSH, for example a TSH of five to nine milliunits per liter, is central hypothyroidism until proven otherwise. Be aware that neonatal reference limits for both hormones run higher in the first two to three weeks of life, that acute illness can lower thyroxine in the so-called sick euthyroid pattern, and that thyroxine binding globulin deficiency lowers total thyroxine but not free thyroxine and needs no treatment.

Cortisol is harder. The newborn has no mature day and night rhythm of cortisol until about six months of age, so the idea of an eight in the morning sample means little. Bulan and colleagues reported in 2023 that, in term infants under thirty days, a random cortisol below five point six micrograms per deciliter, which is about 155 nanomoles per liter, was one hundred percent specific for central adrenal insufficiency, but only about forty-three percent sensitive. That means a low value rules the diagnosis in, but a normal value rules nothing out. The best sample is the critical sample, taken at the moment of hypoglycemia before glucose is given, which should include cortisol, growth hormone, insulin, beta hydroxybutyrate and free fatty acids. When the infant is stable, a low dose corticotropin stimulation test, using one microgram, is the dynamic test for central insufficiency. A brain and pituitary magnetic resonance scan looks for an absent septum pellucidum, optic nerve hypoplasia, a small anterior pituitary, an interrupted stalk and an ectopic posterior pituitary, which may not be visible before about two months of

age. An eye examination completes the work-up.

Part six. Treatment, outcome, and primary hypothyroidism.

The single most important treatment rule is the order: hydrocortisone before levothyroxine. Thyroid hormone speeds up the clearance of cortisol. In an infant whose adrenal reserve is already low, starting levothyroxine first can tip the child into adrenal crisis. The 2021 European congenital hypothyroidism consensus is explicit on this point: in central hypothyroidism, assess the adrenal axis and start hydrocortisone before levothyroxine. In practice, sample first, then give physiological hydrocortisone replacement; the Buenos Aires series used seventeen milligrams per square meter per day. Then start levothyroxine, at least ten micrograms per kilogram per day if free thyroxine is below five picomoles per liter, otherwise five to ten. Monitor treatment by free thyroxine, never by TSH, because the pituitary cannot produce a meaningful TSH. Growth hormone is added for persistent hypoglycemia or confirmed growth hormone deficiency. Two practical details matter in the cholestatic infant. First, impaired fat and drug absorption may mean higher oral doses are needed. Second, families must be taught sick day rules and given an emergency hydrocortisone injection before discharge, and steroid cover must be given for any liver biopsy or surgery.

The liver outcome is good. Every series reports full biochemical recovery after replacement. The median time to resolution was sixty-five days in Buenos Aires, a mean of about nine point seven weeks, roughly sixty-eight days, in Thailand, about eighty-eight days in the German series of Binder and colleagues as cited by later authors, and a median of six months in London, while in the French cohort cholestasis resolved within nine months. A Japanese report documented normal liver histology on repeat biopsy. Aminotransferases can lag behind the bilirubin, and a few historical cases described fibrosis, but no verified liver transplant has been reported for endocrine cholestasis alone. The danger to life lies elsewhere. Brodsky and colleagues described sudden death in five patients with septo-optic dysplasia, attributed to adrenal crisis, which is why the endocrine diagnosis matters more than the liver diagnosis.

Where does primary congenital hypothyroidism fit? Mostly, it causes prolonged unconjugated jaundice, more often when thyroxine is very low, together with constipation, a large tongue and an umbilical hernia. Conjugated cholestasis with a raised GGT has been described in case reports and cleared within weeks of levothyroxine, but when a conjugated picture appears, it is wise to look for a second cause. When the screen is reliable, repeat thyroid testing for prolonged unconjugated jaundice rarely adds anything: Ogundele and Waterson reported in 2010 that 639 thyroid tests in one British unit, most done for prolonged jaundice, found no thyroid diagnosis at all. So the thyroid test that matters in the cholestatic infant is not the one aimed at the thyroid gland. It is free thyroxine with TSH, aimed at the pituitary, together with a cortisol.

Part seven. What the guidelines say.

On thyroid testing, North America and Europe are aligned, because they share one document: the 2017 NASPGHAN and ESPGHAN guideline puts TSH and thyroxine in the first tier only when the newborn screen is not available, and in the second tier otherwise. The British National Institute for Health and Care Excellence guideline on neonatal jaundice asks only that routine metabolic screening, including screening for congenital hypothyroidism, has been performed. On cortisol, the documents are again aligned: it is a second tier test in the joint guideline, and it is not listed in the British prolonged jaundice pathway.

The differences appear when we leave hepatology. The detection of central hypothyroidism differs materially between countries, because the Dutch thyroxine based screen finds it and the TSH based screens of most American states and the United Kingdom do not. The rule of hydrocortisone before levothyroxine is explicit in the 2021 European endocrine consensus but appears nowhere in the hepatology guideline. The adult societies, the American Association for the Study of Liver Diseases and the European Association for the Study of the Liver, have no neonatal cholestasis guideline at all. The most material gap is that no hepatology document links the combination of hypoglycemia and cholestasis to a same day cortisol, although that pairing is the strongest clue we have.

Part eight. Traps and pearls.

The first trap is the reassuring normal screen, which hides central hypothyroidism and leaves free thyroxine unmeasured. The second is the assumption that a low GGT means a genetic transporter disease, so that a panel is sent and the cortisol is not, and weeks are lost waiting. The third is the transitional sugar, when a day two hypoglycemia is forgotten by week four and no critical sample is ever taken. The fourth is the wrong treatment order, when levothyroxine is started first, or a biopsy or operation is done without steroid cover. And one rule sits above all the others: an endocrine answer never closes the biliary atresia question. If the stools are pale or the GGT is high and rising, finish the hepatology work-up on schedule while the hormones are being replaced.

The pearls are worth repeating. Endocrine disease explains about two percent of infant cholestasis, but up to four in ten neonatal hypopituitary infants are cholestatic. Cortisol deficiency is the driver: the lower the cortisol, the more the jaundice, and isolated cortisol deficiency is enough. Hypoglycemia was present in fourteen of sixteen and in eight of eight infants, and it is the best single bedside clue. The GGT is normal in about half of cases. A normal TSH based newborn screen does not exclude central hypothyroidism. A random cortisol below five point six micrograms per deciliter is fully specific but only forty-three percent sensitive. Hydrocortisone comes first and levothyroxine second. And the cholestasis resolves within about two to six months of replacement, while the risk to life is adrenal crisis.

Part nine. What to do differently in clinic.

Return to the two infants at the start. The three-week-old boy with a direct bilirubin of five, a GGT of sixty, a micropenis, wandering eyes and a forgotten day two hypoglycemia has septo-optic dysplasia with multiple pituitary deficiency until proven otherwise. Measure cortisol, free thyroxine and TSH today, give hydrocortisone and then levothyroxine, and arrange a pituitary magnetic resonance scan and an eye examination. Expect the bilirubin to normalize over two to three months. The seven-week-old girl with pale-looking stools, recurrent hypoglycemia and a normal GGT may have isolated cortisol deficiency, but she may also have progressive familial intrahepatic cholestasis or even biliary atresia. Take a critical sample at the next low sugar, measure cortisol, keep the atresia pathway on time, and do not biopsy her without steroid cover. An infant with a free thyroxine of eight picomoles per liter and a TSH of six, whose TSH based screen was normal, has central hypothyroidism and needs an adrenal assessment or hydrocortisone cover before levothyroxine. A thriving three-week-old with unconjugated jaundice and a normal screen most likely has breast milk jaundice, not a thyroid problem, and needs a conjugated bilirubin and reassurance. And the child with known hypopituitarism who is still cholestatic eight weeks into replacement is following the expected slow course; check the doses for malabsorption, follow the liver tests monthly, and expect recovery by three to nine months.

The whole chapter can be held in two sentences. In a cholestatic infant, look at the sugar, the genitals and the eyes, and measure free thyroxine with TSH and a cortisol early, whatever the GGT and whatever the newborn screen said. If the pituitary has failed, give hydrocortisone before levothyroxine, because the liver will recover on its own, but the adrenal crisis will not wait.

Questions to test yourself.

Question one. Which bile transporters are reduced in the livers of infants with congenital hypopituitarism? Answer: the canalicular bile salt export pump and MDR3, with MRP2 reduced in some, which explains why endocrine cholestasis can look like a genetic transporter disease.

Question two. What share of infant cholestasis is endocrine, and what share of neonatal hypopituitary infants are cholestatic? Answer: about two percent of cholestatic infants in the 2015 systematic review, but between six and forty-two percent of infants in pituitary stalk interruption series.

Question three. What is the single most consistent bedside clue? Answer: hypoglycemia, present in fourteen of sixteen infants in the Buenos Aires series and in all eight in the Thai series.

Question four. Why does a normal newborn screen not exclude central hypothyroidism? Answer: because most programs screen on TSH, and in central hypothyroidism the TSH is low, normal or only mildly raised while free thyroxine is low; in the Netherlands, central hypothyroidism is found in about one in sixteen thousand infants by thyroxine based screening.

Question five. How should a random neonatal cortisol below five point six micrograms per deciliter be interpreted? Answer: it is one hundred percent specific but only about forty-three percent sensitive for central adrenal insufficiency, so a low value rules the diagnosis in and a normal value rules nothing out.

Question six. In what order should hormones be replaced, and why? Answer: hydrocortisone first and levothyroxine second, because thyroid hormone increases cortisol clearance and can precipitate adrenal crisis; treatment is then monitored by free thyroxine rather than TSH.

Question seven. How long does endocrine cholestasis take to resolve after replacement? Answer: about two to three months in most series, with a median of sixty-five days in Buenos Aires and a median of six months in London.

End of chapter.